

ECON 304 – Intermediate Macroeconomics

Problem Set 2

Instructor: Sean Callahan | Fall 2026 | 50 points

1 Keynesian Multiplier**Points: 10**

Given the equilibrium condition $Y = C + I + G$, where $C = c_0 + c_1(Y - T)$, $I = \bar{I}$ and $T = G = \bar{G}$:

Question 1.1

Derive the equilibrium output Y^* in terms of the parameters of the model – i.e., c_0 , c_1 , $\bar{I}$, and $\bar{G}$. Include a graph to emphasize the equilibrium level of output.

Question 1.2

Take the derivative of the equilibrium output Y^* with respect to each of the parameters.

Question 1.3

Discuss the behavior of $\frac{dY^*}{dc_1}$. Be sure to address what must be true for the economic system to be stable (identify the mathematical condition for stability and interpret).

Question 1.4

Suppose the federal government cuts taxes by 30% ($T_2 = 0.7T_1$), by how much does the equilibrium output change?

2 IS-LM

Points: 15

Question 2.1

Suppose aggregate demand is given by $Z = c_0 + c_1(Y - T) + \bar{I} + G$ where the government runs a balanced budget. Derive the investment-saving (IS) relation. What does the IS relation represent? Explain.

Question 2.2

Consider the financial market with money demand $\frac{M^D}{P} = L(Y, i) = l_0 Y - l_1 i$ and money supply $\frac{M^S}{P} = \frac{\bar{M}}{P}$. Derive the liquidity-money (LM) relation. What does the LM relation represent? Explain.

Question 2.3

Use the following values to solve for the equilibrium level of output Y^* and the equilibrium nominal interest rate i^* .

Given parameters: $P = 1$, $l_0 = 5$, $l_1 = 2$, $\bar{M} = 490$, $c_0 = 8$, $c_1 = 0.8$, $T = 10$, $\bar{I} = 10$.

3 Response to the Great Recession

Points: 25

Use the equations from the previous questions (2.1 and 2.2) and assume $I(Y, i) = 0.1Y - i$ to illustrate changes to the economic system and your policy response. You will receive a zero if instructions are not followed.

Question 3.1

In 2007–2008, falling house prices caused aggregate demand to fall and the financial system teetered on the verge of collapse. For simplicity, let's assume this is represented by a decline in autonomous expenditure $c'_0 = 1$. Estimate the new equilibrium, graph the Keynesian Cross, and estimate the change in output through the Keynesian multiplier.

Question 3.2

You are the chair of the Federal Reserve, what is your immediate response to the crisis? Illustrate your open market operations by making a change to one of the equations previously used. What is the new equilibrium and provide intuition for why it has shifted.

Question 3.3

As chair of the Fed, are you satisfied with the results of your monetary operations? Are you concerned about the efficacy of future monetary policy?

Question 3.4

How should the federal government respond to this crisis? Illustrate graphically and algebraically.